

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Lost and found portal (web based item recovery sytem)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work being presented in the project report entitled “**Lost & Found Portal – Web-Based Item Recovery System**” in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering), submitted to the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Dr. Monika Aggarwal..

The matter presented in this project report has not been submitted to any other university or institute for the award of any degree.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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ABSTRACT

In today's fast-moving environment, people frequently lose personal belongings such as wallets, mobile phones, documents, ID cards, bags, and electronic devices in campuses, offices, and public places. Traditional methods of managing lost and found items mainly rely on manual reporting, notice boards, or social media posts, which are often unorganized, time-consuming, and inefficient. Due to the lack of a centralized system, users face difficulties in identifying item owners, communicating effectively, and recovering lost belongings.

Lost & Found Portal – Web-Based Item Recovery System is a full-stack web application developed to solve these challenges using modern web technologies. The platform provides a centralized system where users can report lost items, post found items, upload images, search items using category and location filters, and communicate securely through a messaging system.

The application includes secure user authentication, item status tracking, admin approval, and responsive dashboard management. Users can easily browse, filter, and manage item records, improving transparency and increasing the chances of successful item recovery.

The system is developed using the MERN stack, including MongoDB for database management, Express.js and Node.js for backend operations, and React.js for frontend development. The platform demonstrates how modern web technologies can be effectively used to solve real-world item management problems through a secure, scalable, and user-friendly digital solution.

1. INTRODUCTION

The increasing use of digital technologies and growing movement of people across campuses, workplaces, and public spaces has led to a rise in cases of lost and misplaced items. People often lose valuable belongings such as mobile phones, wallets, identity cards, documents, bags, and accessories during daily activities. Recovering such items becomes difficult because there is usually no proper centralized system to manage lost and found information efficiently.

Traditional methods of managing lost and found items generally depend on notice boards, verbal communication, or social media posts. These methods are often unorganized, unreliable, and time-consuming. Users face problems such as lack of proper communication, duplicate reports, fake claims, and difficulty in identifying the real owner of an item.

With the advancement of modern web technologies, digital solutions can now be developed to simplify item management and communication processes. A centralized web-based system can help users quickly report, search, and recover lost belongings while maintaining proper records and secure communication.

Lost & Found Portal is developed as a smart web-based platform that provides users with features such as item posting, image upload, category-based filtering, messaging, item tracking, and admin verification. The system aims to improve item recovery efficiency and replace traditional manual methods with an organized digital solution.

1.1 Background of the Study

The rapid growth of educational institutions, workplaces, shopping centers, and public transport systems has increased the chances of people losing personal belongings during daily activities. Many individuals struggle to recover lost items because existing systems are mostly manual and lack proper organization.

Manual systems create several challenges such as delayed communication, fake ownership claims, missing information, and low recovery rates. People often depend on physical notice boards or informal communication channels, which are not efficient for managing large numbers of lost and found records.

Modern web-based systems provide opportunities to create centralized digital platforms that simplify information management and improve communication between users. By integrating features such as secure authentication, image upload, search filters, and messaging systems, item recovery processes can become faster and more reliable.

1.2 Problem Statement

In the current environment, there is no proper centralized system available for managing lost and found items efficiently. Traditional methods such as notice boards, verbal announcements, and social media posts are unorganized and difficult to manage.

Users face problems such as poor communication, difficulty in identifying item owners, fake claims, and low recovery rates. Existing methods also lack proper tracking and secure management of item information.

There is a need for a secure, organized, and user-friendly web-based platform that allows users to report lost items, post found items, search records efficiently, and communicate safely to improve item recovery.

1.3 Objectives of the Project

The primary objectives of the Lost & Found Portal project are as follows:

- To develop a centralized web-based lost and found management system.
- To simplify the process of reporting and searching for lost items.
- To provide secure user authentication and authorization.
- To implement category and location-based item filtering.
- To enable communication between users through a messaging system.
- To improve item recovery rates through organized digital management.

1.4 Scope of the Project

The scope of the Lost & Found Portal includes the development of a web-based platform for managing lost and found items efficiently. The system supports user registration, item posting, image upload, search functionality, secure messaging, and admin approval.

The platform is suitable for campuses, offices, public places, and organizations where item recovery management is required.

Future enhancements may include mobile application support, AI-based item matching, Google Maps integration, notification systems, and cloud deployment.

2. METHODOLOGY

The development of Lost & Found Portal followed a structured software development methodology to ensure proper planning, implementation, testing, and deployment of all modules.

The first phase involved requirement analysis, where problems related to traditional lost and found management systems were studied. Based on this analysis, system objectives and core features were identified.

A modular client-server architecture was adopted for development. The frontend was developed using React.js to provide an interactive and responsive user interface. The backend was developed using Node.js and Express.js to handle APIs, authentication, and business logic. MongoDB was used as the database management system for storing user, item, and messaging data.

The workflow begins with user registration and secure authentication. Users can then post lost or found items with descriptions and images. Admin verification ensures authenticity before items are publicly displayed.

Testing and debugging were performed throughout the development process to ensure smooth integration between frontend, backend, and database systems.

3. TOOLS AND TECHNOLOGIES

The development of the Lost & Found Portal – Web-Based Item Recovery System involved the use of modern web development technologies, database systems, authentication tools, and development utilities to create a secure, scalable, and user-friendly application. The selected technologies were chosen based on performance, flexibility, ease of integration, and suitability for full-stack web application development.

3.1 Frontend Technologies

React.js:

React.js was used for developing the frontend of the application. It is a popular JavaScript library used for building responsive and interactive user interfaces. React follows a component-based architecture, which allows developers to create reusable UI components and manage application states efficiently. It helped in creating dynamic pages such as login forms, dashboards, item posting pages, and messaging interfaces. React also improves application performance by updating only the required parts of the webpage using the virtual DOM concept.

tailwind CSS:

Tailwind CSS and CSS were used for designing and styling the user interface of the application. tailwind CSS provides utility-based classes that help in creating responsive and modern designs quickly without writing large amounts of custom CSS code. It was used to design forms, buttons, cards, dashboards, and navigation components. CSS was additionally used for custom styling and layout adjustments to improve the visual appearance and user experience of the platform.

Axios:

Axios was used as an HTTP client for communication between the frontend and backend. It enabled smooth API request handling for user authentication, data retrieval, and progress tracking.

Vite:

Vite was used as the frontend build tool and development server. It provides faster project initialization, hot module replacement, and optimized build performance compared to traditional frontend tooling.

3.2 Backend Technologies

Node.js:

Node.js was used for server-side application development. It provides an event-driven runtime environment suitable for building scalable backend systems and handling asynchronous operations efficiently.

Express.js:

Express.js was used as the backend web application framework. It simplified API creation, request handling, routing, middleware integration, and overall backend architecture development.

JSON Web Token (JWT):

JWT was used for implementing secure user authentication and authorization. It ensured secure session management and protected access to restricted application routes.

bcryptjs:

bcryptjs was used for hashing user passwords before storing them in the database. This improved application security by preventing direct password exposure.

3.3 Database Technology

MongoDB:

MongoDB was used as the primary database management system for storing application data. It is a NoSQL database that stores data in flexible JSON-like documents, making it suitable for handling dynamic and scalable web applications. MongoDB was used to store user information, item records, categories, messages, and item status data. Its flexibility and high performance made it suitable for managing large amounts of application data efficiently.

Mongoose:

Mongoose was used as an Object Data Modeling (ODM) library for MongoDB. It helped in defining database schemas, validating data, and simplifying database operations within the backend application. Mongoose provided structured models for users, items, categories, and messages, improving data consistency and application maintainability.

3.4 Development Tools

Visual Studio Code:

Visual Studio Code was used as the primary code editor for frontend and backend development. It provides features such as syntax highlighting, debugging tools, extension support, integrated terminal, and Git integration. These features improved coding efficiency and simplified project management during development.

Git and GitHub:

Git was used for version control to track changes made during development. GitHub was used as the online repository hosting platform for storing source code and collaborating on the project. These tools helped in managing project versions, maintaining backups, and enabling team collaboration efficiently.

Postman:

Postman was used for API testing and debugging. It helped in sending API requests, testing backend routes, validating server responses, and debugging authentication functionality. Postman improved backend testing efficiency and ensured smooth communication between frontend and backend services.

MongoDB Compass:

MongoDB Compass was used for visually managing and inspecting database records during development and testing.

4. IMPLEMENTATION

The implementation of the Lost & Found Portal – Web-Based Item Recovery System was carried out using a modular full-stack development approach. Different modules of the application were developed separately and later integrated into a single system. The platform combines frontend user interaction, backend processing, database management, authentication, and communication features to provide a complete lost and found management solution.

4.1 User Authentication Module

The User Authentication Module was implemented to provide secure user registration and login functionality. Users can create accounts by entering details such as name, email, and password. Passwords are encrypted using bcrypt before being stored in the database, ensuring secure credential management and preventing direct password exposure.

JSON Web Tokens (JWT) were used for secure authentication and authorization. After successful login, a token is generated and provided to the user, allowing access to protected routes and personalized dashboard features. This module ensures that only authenticated users can perform actions such as posting items, sending messages, and managing personal information.

The authentication system improves overall platform security and maintains reliable user session management.

4.2 Item Management Module

The Item Management Module was implemented to manage lost and found item records within the system. Users can post lost items or found items by providing details such as item title, description, category, location, date, and images.

The module also supports updating and deleting item records whenever required. Uploaded images help users identify items more accurately and improve the chances of successful recovery.

Each item includes a status field such as Open, Claimed, or Closed, which helps track the recovery progress of items. Backend APIs handle item storage, retrieval, updates, and deletion operations efficiently.

This module forms the core functionality of the Lost & Found Portal system.

4.3 Category Management Module

The Category Management Module was developed to organize lost and found items into different categories for easier searching and filtering. Categories such as electronics, documents, wallets, bags, accessories, and others help users quickly locate relevant items.

Admin users can add, update, or delete categories through the management panel. The frontend provides filtering functionality that allows users to search items based on selected categories and locations.

This module improves system organization, user experience, and item search efficiency.

4.4 Messaging Module

The Messaging Module was implemented to provide secure communication between users regarding lost and found items. Interested users can contact the item owner directly through the messaging feature available on the platform.

Messages are stored securely in the database and linked to specific users and item records. This module helps users coordinate item recovery, verify ownership details, and exchange additional information safely.

The messaging system improves communication efficiency and increases the possibility of successful item recovery.

4.5 Admin Module

The Admin Module was implemented to manage and monitor overall platform activities. Administrators have access to special functionalities such as item approval, category management, user management, and system monitoring.

Before items are publicly displayed, admin verification is performed to reduce fake or inappropriate posts. The admin panel also allows administrators to manage user accounts and monitor reported activities within the platform.

This module improves platform reliability, transparency, and overall system control.

4.6 Database Implementation

MongoDB was used as the primary database management system for storing all application data. Separate collections were created for users, items, categories, and messages.

Mongoose schemas were implemented to define structured models and validate data before storing it in the database. The database stores information such as user details, item descriptions, uploaded images, categories, item statuses, and communication records.

Efficient database integration enabled smooth storage, retrieval, and management of application data while maintaining scalability and performance.

4.7 API Integration

RESTful APIs were developed using Express.js to manage communication between the frontend and backend of the application. APIs handle operations such as user authentication, item posting, image uploading, category filtering, messaging, and admin approval.

Axios was used on the frontend for sending API requests and receiving responses from the backend server. Proper API integration ensured smooth data exchange and real-time interaction between different components of the system.

The API-based architecture improved modularity, maintainability, and scalability of the application.

4.8 Testing and Integration

Testing and debugging were performed throughout the development process to ensure proper functionality of all modules. Individual modules such as authentication, item management, messaging, and admin approval were tested independently before integration.

Frontend responsiveness, backend API functionality, database connectivity, and authentication workflows were validated carefully. Integrated testing ensured smooth communication between frontend, backend, and database systems.

The system was successfully integrated and tested to provide stable performance, secure operations, and a responsive user experience across all functionalities.

5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

The Lost & Found Portal – Web-Based Item Recovery System was successfully developed and tested with all core functionalities operating efficiently. The project demonstrated the effective use of full-stack web technologies in solving real-world problems related to lost and found item management.

One of the major outcomes of the project was the successful implementation of a centralized digital platform for managing lost and found items. Users were able to report lost items, post found items, upload images, and search item records through a single integrated system. This significantly reduced the confusion and inefficiency commonly associated with traditional manual methods such as notice boards and social media posts.

The item posting and image upload features functioned effectively by allowing users to provide detailed descriptions and photographs of items. The uploaded images improved item identification accuracy and increased the chances of successful item recovery. Users could easily manage their posted items through the dashboard interface.

The category and location-based filtering system successfully improved search efficiency within the platform. Users were able to search and filter items based on categories such as electronics, wallets, documents, bags, and accessories, making the item retrieval process faster and more organized. Location-based searching also helped users identify items found in specific areas.

The messaging system was another important achievement of the project. It enabled secure communication between users regarding item ownership and recovery. Users could exchange information safely without exposing sensitive personal details publicly. This feature improved coordination and simplified the process of verifying ownership.

The admin approval system played a significant role in improving platform reliability and transparency. Before being displayed publicly, item posts were verified by administrators to reduce fake or misleading entries. This improved the trustworthiness and authenticity of the platform.

The secure authentication system using JWT and bcrypt successfully protected user accounts and restricted unauthorized access to sensitive features. User data and login credentials were stored securely, ensuring privacy and security throughout the application.

From a technical perspective, the MERN stack architecture provided scalability, modularity, and efficient full-stack application development. Frontend responsiveness, backend API communication, database integration, and authentication workflows operated smoothly and provided a stable user experience.

Testing and integration of all modules were successfully completed. Functional testing ensured that each module worked independently, while integrated testing validated smooth communication between frontend, backend, and database systems. The application performed reliably across all implemented features.

Overall, the project successfully demonstrated how modern web technologies can be effectively used to solve real-world item management problems. The Lost & Found Portal provides a practical, secure, scalable, and user-friendly solution that improves communication, transparency, and item recovery efficiency in campuses, workplaces, and public environments.

6. CONCLUSION AND FUTURE SCOPE

The Lost & Found Portal was successfully developed as a secure and efficient web-based item recovery system designed to simplify lost and found item management.

The platform effectively replaces traditional manual methods such as notice boards and paper records with a centralized digital solution. Features such as item posting, image upload, secure authentication, messaging, category filtering, and admin approval improve transparency, communication, and item recovery efficiency.

The project demonstrates how modern web technologies like React.js, Node.js, Express.js, and MongoDB can be integrated to build scalable and practical real-world applications.

Future enhancements can further improve the capabilities of the system. Integration with Google Maps can provide accurate location tracking for lost and found items. AI-based item matching systems can automatically suggest matching items based on category, image, and location data.

A mobile application version can improve accessibility and convenience for users. Real-time notifications and email alerts can further enhance communication and item tracking.

Additional improvements may include multi-language support, cloud deployment, QR-based item verification, and advanced analytics for item management.

Overall, the Lost & Found Portal represents a practical, scalable, and future-ready digital solution for efficient lost and found item recovery management.

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- [10] bcryptjs Package [Documentation](#).
- [11] Visual Studio Code [Documentation](#).
- [12] Postman [Documentation](#).

APPENDICES

Appendix A: User Login Interface Screenshot

Create Account

Start your recovery journey today

Nickname *



Full Name *



Email *

 email@example.com

Password *



Sign Up

OR

Already part of our community?

Sign In



Sign In

Access your recovery dashboard

Email Address



kumarpritampyare786@gmail.com

Password



••••••••



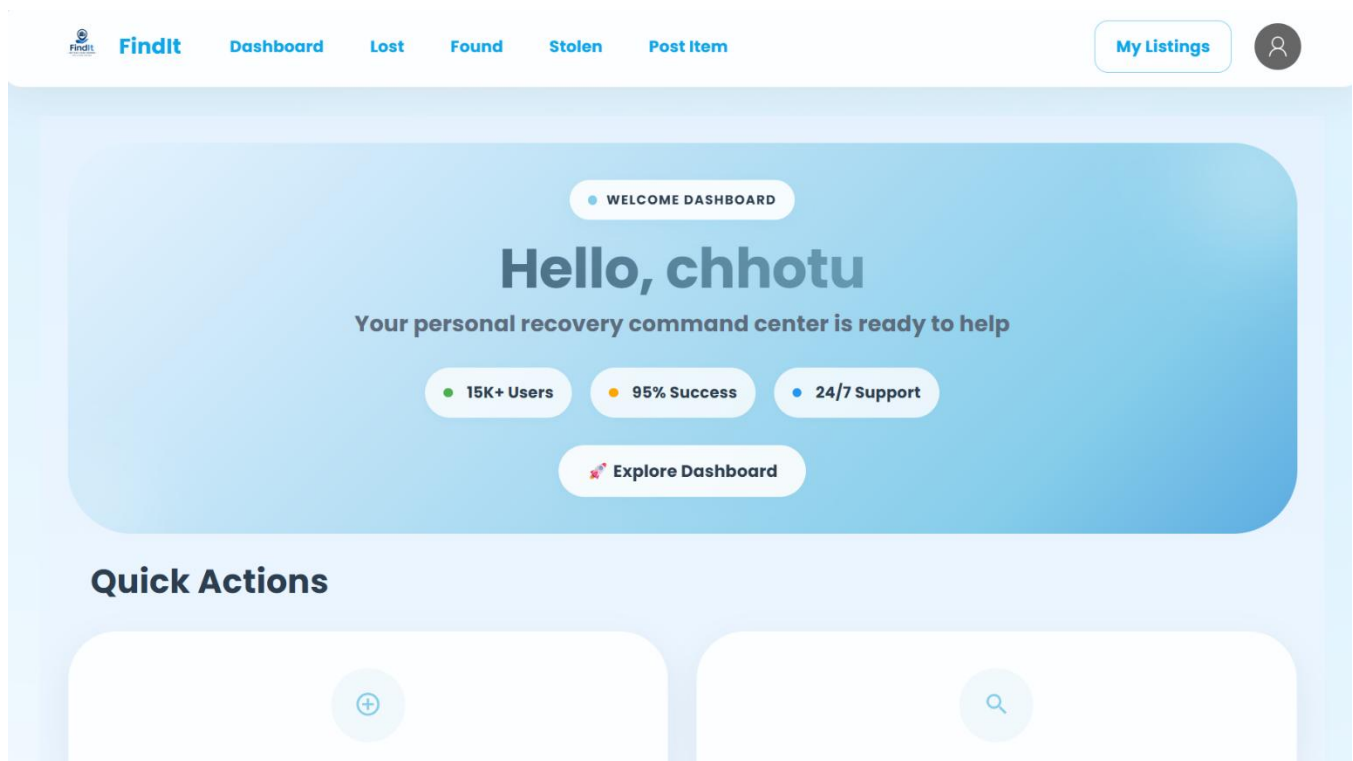
→ Sign In

OR

New to FindIt? Start your recovery journey

Create Account

Appendix B: Dashboard Interface Screenshot



Appendix C: Post Item for recovery Selection Interface Screenshot

Choose Images (0/3)

• Upload clear photos of the lost/found item • Maximum file size: 5MB per image • Supported formats: JPG, PNG, GIF, WebP • Maximum 3 images per listing • Images are stored locally in the database

Item Information

Item Name *

Item Type

Description *

Location *

Date *
dd-mm-yyyy

Contact Number *

Create Listing

Appendix C: In My list where list create Interface Screenshot

Findit

Dashboard

Lost

Found

Stolen

Post Item

My Listings

Welcome back, chhotu

Manage all of your lost, found, and stolen listings from one dashboard.

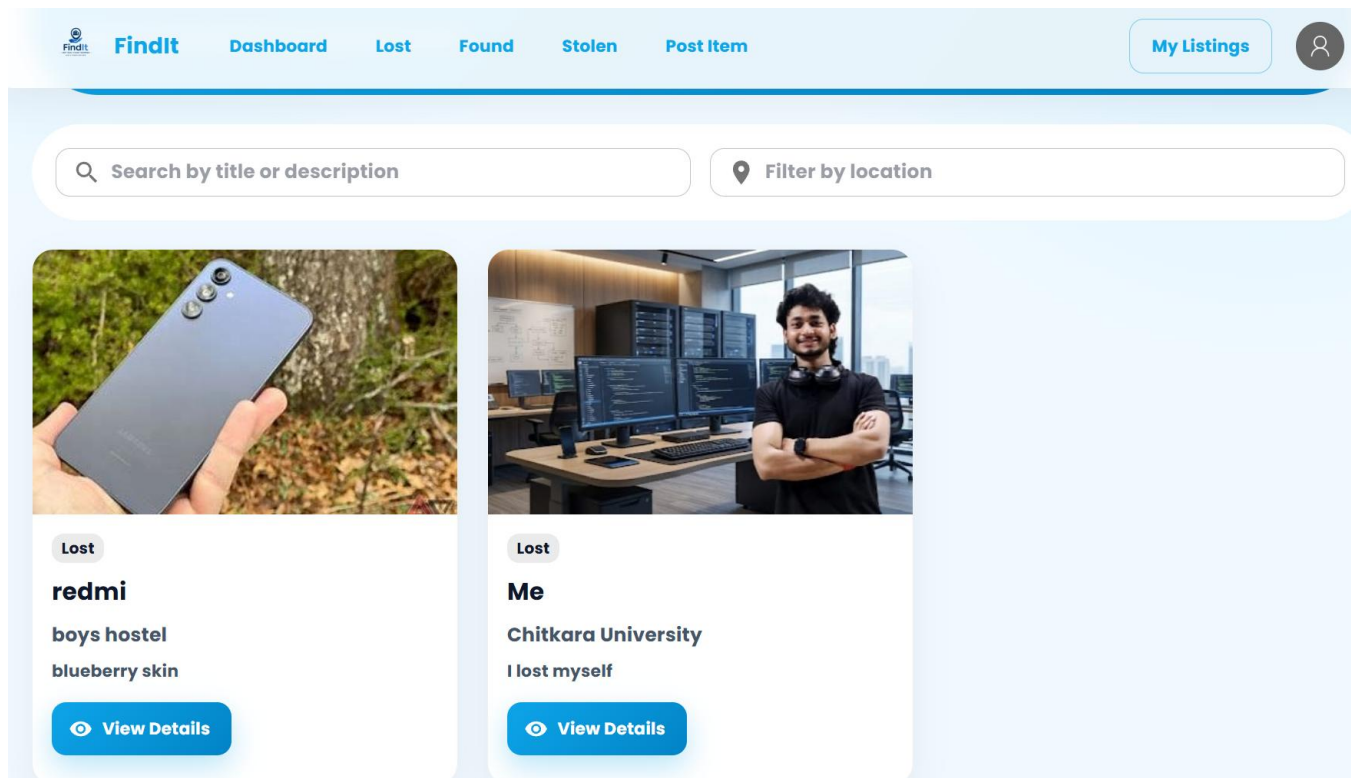
+ Post New Item

You have not posted any items yet.

Start by creating your first listing.

Create Listing

Appendix E: Lost Item List Overview



This project demonstrates the effective integration of full-stack web development technologies in creating a practical and efficient item recovery management solution.

***** End of Report *****